SHAYINI SIDDARDHA

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PROFESSIONAL SUMMARY

AI/ML Engineer specializing in Local LLMs, Agentic AI, Computer Vision, and Retrieval-Augmented Generation (RAG). Creator of FastAgent, an open-source hardware-adaptive local LLM inference orchestrator designed for CPU-only edge devices. Experience building production-oriented AI systems, contributing to open-source projects, and publishing research in IEEE Xplore.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C, C++, HTML/CSS
AI/ML: LLMs, Agentic AI, RAG, LangChain, TensorFlow, scikit-learn, NLP, Prompt Engineering, YOLOv12n, BoT-SORT, Kalman Filter
Frameworks & Tools: FastAPI, Flask, Streamlit, Git, GitHub, Ollama, llama.cpp, SQLite, LanceDB, REST APIs

WORK EXPERIENCE

AI Engineer Intern | Infosys Springboard — Feb 2025 – Apr 2025

- Developed an electricity demand and price forecasting system using Facebook Prophet with ~95% prediction accuracy.
- Built a Flask backend and interactive web interface for real-time forecasting.

AI Engineer Intern | YBI Foundation — Dec 2024 – Jun 2025

- Built an Air Quality Prediction System using Random Forest Regression on Indian city datasets.
- Achieved ~96% AQI category accuracy through optimized preprocessing and model training.

PROJECTS

FastAgent *(Open Source • Research Published)* | GitHub | Publication — Jun 2026 – Sep 2026

- Developed a hardware-adaptive local LLM inference orchestrator using **llama.cpp**, prefix caching, KV-cache quantization, grammar-constrained decoding, and adaptive compute control.
- Reduced steady-state prompt latency from **~3500 ms to ~73 ms** on CPU-only 8 GB systems while preventing OOM failures.

WildGuardAI *(IEEE Published)* | GitHub | Publication — Aug 2025 – Jun 2026

- Built Built a wildlife conflict prediction system using YOLOv12n, BoT-SORT, Kalman Filter, Flask, and SQLite.
- Published research in **IEEE Xplore** for real-time wildlife monitoring and alert generation.

JivanMitra | GitHub — Jul 2025 – Mar 2026

- Developed an offline-first AI assistant using LangChain, RAG, and Agentic AI for multilingual elderly assistance.
- Designed for privacy-first deployment with local inference and contextual memory.

Open Source Contributor — sklearn-genetic-opt | GitHub PR #178 — Sep 2025 – Mar 2026

- Contributed documentation improvements for parallel processing in GASearchCV.
- Pull Request **#178** merged, resolving Issue **#152**.

EDUCATION

Bachelor of Technology – Computer Science & Engineering (AI & ML) — 2022 – 2026
Vignan Institute of Technology and Science, Hyderabad | CGPA: 7.72/10.00

CERTIFICATIONS & ASSESSMENTS

- **TCS iON NQT (IT):** Qualified – Dec 2025 | Strong performance in Python & problem-solving
- **NPTEL:** Programming with Generative AI – Nov 2025
- **Oracle:** Agentic AI Certified Foundations Associate (Jul 2026), Oracle Certified Foundations Associate (Oct 2025)
- **Coursera:** Cloud Machine Learning Engineering & MLOps – Dec 2024
- **Infosys Springboard:** DSA, AI Primer, Principles of Generative AI, Generative Models for Developers